

The Drill Decks

Chapter 1 · Theorems 1–16 · print on card stock, cut on the dashed lines

What is in the box

G · The Givens (36 cards). All twenty-four definitions, five postulates, five common notions and two further principles. The fourteen marked ★ do nearly all the work in Theorems 1–16 — separate those out and drill them first, to reflex speed.

T · The Theorems (16 cards). Number, title, statement, way of demonstrating, how it is proved, and what each one rests on.

S · The Ways of Demonstrating (5 cards). The five ways a theorem is demonstrated in Chapter 1.

O · The Operations (9 cards). The nine recurring actions, and what permits each one.

The letter in the bottom corner of each card tells you which deck it belongs to after everything has been shuffled together — which it will be.

How to run them

Cite-It (deck G). Read the cue at the foot of the card; they answer with the label in the band. Twenty items in two minutes, in unison, with a signal. Then reverse it: read the label, they give the statement.

Two-way deck (deck T). Read the number; they give the title and statement. Then read the statement; they give the number. The second direction is the one that matters and the one most decks never train.

Which way of demonstrating? (decks T and S). Read only the statement — no diagram, no number. They name the way, and how it is proved. Mix the ways deliberately; never five of the same in a row.

Which operation? (deck O). Describe a figure and a goal; they name the operation it calls for.

One rule for all four. Ask · pause · signal · respond. The pause is the whole mechanism: signal too early and the faster boy answers while the younger one echoes.

DEFINITION 1

Solid

A solid is whatever has length, width, and depth.

Cite-it cue: "Length, width and depth."

G

DEFINITION 2

Surface

A solid stops at its surface; a surface has length and width, but no depth.

Cite-it cue: "Where a solid stops."

G

DEFINITION 3

Line

When a surface comes to an end it stops at a line; a line has length, but no width or depth.

Cite-it cue: "Where a surface stops."

G

DEFINITION 4

Point

A point is a magnitude with no parts. When a line comes to an end it stops at a point; so a point has no length, no width and no depth. Though it has no shape or size, it does have one positive feature: its location.

Cite-it cue: "A magnitude with no parts; where a line stops."

G

DEFINITION 5

Straight line

A straight line is a perfectly uniform line: every part is the same shape as every other part.

Cite-it cue: "A line the same shape all along."

G

DEFINITION 6

Plane (flat surface)

A flat surface is a perfectly uniform surface — the same shape all over and on both sides.

Cite-it cue: "A surface the same shape all over."

G

DEFINITION 7

Plane angle

When two distinct lines in a plane meet at a point, the inclination of one to the other is a plane angle. The meeting point is the vertex.

Cite-it cue: "Two lines meeting at a point."

G

DEFINITION 8

Rectilineal angle

A rectilineal angle is an angle formed by two different straight lines.

Cite-it cue: "An angle made by two straight lines."

G

DEFINITION 9

**Right angle**

When one straight line stands on another so that the two adjacent angles formed are equal to each other, each is called a right angle.

Cite-it cue: "Two adjacent angles, equal to each other."

G

DEFINITION 10

**Perpendicular**

A straight line standing on another at right angles is perpendicular to the line on which it stands.

Cite-it cue: "A line standing on another at right angles."

G

DEFINITION 11

Obtuse and acute angles

An obtuse angle is greater than a right angle; an acute angle is less than a right angle.

Cite-it cue: "Greater than a right angle; less than."

G

DEFINITION 12

Supplementary and complementary

Supplementary: they add to two right angles.

Complementary: two acute angles adding to one right angle.

Cite-it cue: "Two angles adding to two rights."

G

DEFINITION 13

Boundary

A boundary of a thing is its limit, or where it stops.

Cite-it cue: "Where a thing stops."

G

DEFINITION 14

Figure

A figure is something contained by its boundaries — which cannot be entered or left without cutting across them.

Cite-it cue: "Contained by its boundaries."

G

DEFINITION 15

Circle

A circle is a plane figure contained by one curved line whose every point is the same distance from a single point inside it.

Cite-it cue: "One curved line; every point the same distance from a point inside." G

DEFINITION 16

Center and circumference

The point inside that is the same distance from every point of the bounding line is the centre; the bounding line is the circumference.

Cite-it cue: "The point inside; the line around." G

G

DEFINITION 17

**Radius and diameter**

Any straight line from the centre of a circle to the circumference is a radius. All radii of a circle are equal. A line through the centre ending at the circumference both ways is a diameter.

Cite-it cue: "Two lines from the centre to the edge."

G

DEFINITION 18

Semicircle

A semicircle is the figure contained by a circle's diameter and the circumference cut off by it.

Cite-it cue: "A diameter and the arc it cuts off."

G

DEFINITION 19

Rectilineal figures

A plane figure contained by straight lines only. Three lines: a triangle. Four: a quadrilateral. More than four: a polygon.

Cite-it cue: "Three straight lines; four; more than four."

G

DEFINITION 20

**Equilateral, isosceles, scalene**

Equilateral: all three sides equal. Isosceles: only two sides equal. Scalene: no equal sides.

Cite-it cue: "All three sides equal; only two; none."

G

DEFINITION 21

Right, obtuse, acute triangles

A right triangle contains a right angle; an obtuse triangle an obtuse angle; an acute triangle has all three acute. The side opposite the right angle is the hypotenuse.

Cite-it cue: "A triangle containing a right angle."

G

DEFINITION 22

Parallel

Two straight lines in one plane that never meet, however far they are extended in either direction.

Cite-it cue: "Two lines in a plane that never meet."

G

DEFINITION 23

Square, rectangle, rhombus, parallelogram

Square: four sides equal, four angles right. Rectangle: four right angles, not all sides equal. Rhombus: four sides equal, no right angles. Parallelogram: two pairs of parallel sides.

Cite-it cue: "Four sides equal and four angles right."

G

DEFINITION 24

Inclined toward each other

Two straight lines cut by a third are inclined toward each other if the two interior angles on one side are together less than the two exterior angles on that side.

Cite-it cue: "Leaning toward each other."

G

POSTULATE 1**Draw a straight line**

A straight line can be drawn from any point to any point.

Cite-it cue: "Join these two points."

G

POSTULATE 2**Extend a straight line**

Any straight line can be extended continuously in a straight line, in either direction, as far as you please.

Cite-it cue: "Produce this line further."

G

POSTULATE 3**Draw a circle**

A circle can be drawn around any point as its centre and with a radius of any given length.

Cite-it cue: "Draw a circle about this point with this radius."

G

POSTULATE 4**Right angles are equal**

All right angles are equal — not only adjacent ones, but those that are not adjacent.

Cite-it cue: "Two right angles here, two right angles there."

G

POSTULATE 5**Inclined lines meet**

Straight lines inclined toward each other eventually meet, when extended far enough.

Cite-it cue: "Two lines leaning toward each other — do they meet?"

G

COMMON NOTION 1**Transitivity of equality**

Things equal to the same thing are also equal to each other.

Cite-it cue: "Each equals the same third thing."

G

COMMON NOTION 2**Adding equals**

When equals are added to equals, the wholes are equal.

Cite-it cue: "Add equals to equals."

G

COMMON NOTION 3**Subtracting equals**

When equals are subtracted from equals, the remainders are equal.

Cite-it cue: "Take equals from equals."

G



Coincidence

Things that can be made to coincide with each other are equal.

Cite-it cue: "Lay one exactly on top of the other."

G

FURTHER PRINCIPLE



Two straight lines cannot enclose a space

Two straight lines cannot cut each other more than once; they cannot enclose a space. This follows from the perfect uniformity of straight lines.

Cite-it cue: "Could two straight lines cut each other twice?"

G

THEOREM 1



How to make an equilateral triangle

Given a straight line AB, construct an equilateral triangle on it so that AB is the base.

Way A · proved by Two circles (Post. 3 + Def. 17), chained by C.N. 1 · rests on — T

THEOREM 3



Base angles of an isosceles triangle

In an isosceles triangle the base angles are equal to each other, and the angles under the base are equal to each other.

Way B · proved by SAS twice, then subtract equals (C.N. 3) · rests on Thm. 2 T



Whole and part

Every whole is greater than any one of its parts.

Cite-it cue: "A whole, and one of its parts."

G

FURTHER PRINCIPLE

Halves of equals are equal

The halves of equal things are equal.

Cite-it cue: "Halves of equal things."

G

THEOREM 2



Side-Angle-Side (SAS)

If in one triangle a side, the next angle, and the next side equal a side, the next angle, and the next side in another triangle, then all corresponding sides and angles are equal, and the areas are equal.

Way B · proved by Superposition (C.N. 4); two lines cannot enclose a space · rests on — T

THEOREM 4



Converse of the isosceles base-angle theorem

If two angles in a triangle are equal to each other, then the sides opposite them are also equal.

Way C · proved by Reductio, landing on C.N. 5 · rests on Thm. 2 T

THEOREM 5

C

The three sides determine the angles — triangles are rigid

In a triangle, the lengths of the three sides determine the three angles: you cannot change any angle while keeping all three side lengths the same.

Way C · proved by Thm. 3 twice, a substitution, then C.N. 5 · rests on Thm. 3

T

THEOREM 6

B

Side-Side-Side (SSS)

If the three sides of one triangle equal the three corresponding sides of another, the corresponding angles are equal and so are the areas.

Way B · proved by Follows from Theorem 5, then C.N. 4 · rests on Thm. 5

T

THEOREM 7

A

How to bisect any rectilineal angle

Given any rectilineal angle ABC, construct the ray that bisects it.

Way A · proved by Equilateral triangle (Thm. 1), verified by SSS · rests on Thms. 1, 6

T

THEOREM 8

A

How to bisect any straight line

Given a straight line AB, construct its midpoint.

Way A · proved by Thm. 1 + Thm. 7, verified by SAS · rests on Thms. 1, 7, 2

T

THEOREM 9

A

Erect a perpendicular from a point on a line

Given a straight line AB and a point P on it, draw a line from P at right angles to AB.

Way A · proved by Thm. 1 + SSS, then Def. 9 and Def. 10 · rests on Thms. 1, 6

T

THEOREM 10

A

Drop a perpendicular from a point to a line

Given a straight line AB and a point P not on it, drop a perpendicular from P to AB.

Way A · proved by Thm. 8 + SSS, then Def. 9 and Def. 10 · rests on Thms. 8, 6

T

THEOREM 11

D

Adjacent angles on a straight line

When one straight line stands on another, the adjacent angles add up to two right angles.

Way D · proved by A constructed perpendicular (Thm. 9), Def. 9, C.N. 1 · rests on Thm. 9

T

THEOREM 12

C

Converse: angles sum to two rights $\Rightarrow$ straight line

If two adjacent angles add up to two right angles, then the outer legs form one straight line.

Way C · proved by Thm. 11 + Post. 4, landing on C.N. 5 · rests on Thm. 11

T

THEOREM 13

D

Vertical angles are equal

When two straight lines cut each other, they make the vertical (opposite) angles equal.

Way D · proved by Thm. 11 twice, then C.N. 3 · rests on Thm. 11

T

THEOREM 14

B

Exterior angle greater than remote interior

If any side of a triangle is extended, the exterior angle is greater than either of the interior and opposite angles.

Way B · proved by Thm. 8 + Thm. 13 + SAS, finishing on C.N. 5 · rests on Thms. 8, 13, 2

T

THEOREM 15

B

Greater side opposite greater angle

In any triangle, a greater side has opposite to it a greater angle.

Way B · proved by Cut off an equal length + Thm. 3 + Thm. 14 + C.N. 5 · rests on Thms. 3, 14

T

THEOREM 16

E

Greater angle opposite greater side

In any triangle, a greater angle has opposite to it a greater side.

Way E · proved by Thm. 3 and Thm. 15, by eliminating two of three cases · rests on Thms. 3, 15

T

WAY A

A

Construct and Verify

SHAPE · Build the figure with circles and lines — then prove that what you built does what you claimed, always by SAS or SSS. TELL · The statement begins “Given ... construct ...”.

Theorems: 1 · 7 · 8 · 9 · 10 · **they will skip:** They build it and stop. Half the demonstration is the verification.

S

WAY B

B

Discover Congruences

SHAPE · Prove two triangles equal by SAS or SSS, then take the corresponding part you were after all along. TELL · The goal is an equality of two segments or two angles.

Theorems: 3 · 6 · 14 — and the second half of every Way A proof · **they will skip:** Naming which triangles, in corresponding order, and which test. “They’re congruent” is not a step.

S

WAY C

C

Reductio ad Absurdum

SHAPE · Assume the contrary; run it forward honestly; land on a whole equal to its own part; declare the absurdity. TELL · The statement is a converse, or asserts an impossibility.

Theorems: 4 · 5 · 12 · **they will skip:** Ruling out the other direction. One case closed is not a trichotomy closed.

S

WAY D

D

Find What Is Equal

SHAPE · Two quantities are each equal to two right angles, or to a common thing. Conclude by C.N. 1; strip a common piece by C.N. 3. TELL · The phrase “two right angles” appears in the statement.

Theorems: 11 · 12 · 13 · **they will skip:** Noticing that the perpendicular is constructed (Thm. 9), not given.

S

Trichotomy Elimination

SHAPE · Exactly three cases are possible — greater, equal, less. Kill two with theorems already proved. The survivor is the conclusion. TELL · The converse of an inequality theorem.

Theorems: 16 · **they will skip:** Attempting a construction. There is none — it is pure logic.

S

OPERATION 1

Draw a circle to get equal lines

Licensed by Post. 3 + Def. 17. The engine of the chapter: a circle hands you equality without measuring.

Used in: 1 · 3 · 4 · 7 · 9 · 10

O

OPERATION 2

Join two points

Licensed by Post. 1. Free, always available, and easy to forget to cite.

Used in: 1 · 3 · 4 · 5 · 7 · 9 · 10 · 14 · 15

O

OPERATION 3

Extend a line past its end

Licensed by Post. 2. Needed whenever an exterior angle or an “under the base” angle appears.

Used in: 12 · 14 — and the extension in 3

O

OPERATION 4

Cut off a length equal to a given one

Licensed by Post. 3 (a circle). Watch which segment you lay it off along — always the greater one.

Used in: 3 · 4 · 15

O

OPERATION 5

Erect the equilateral triangle on a segment

Licensed by Thm. 1. The universal scaffold for the constructions.

Used in: 7 · 8 · 9

O

OPERATION 6

Bisect — an angle, or a line

Licensed by Thm. 7 · Thm. 8. Bisection creates two equal pieces you can feed to SAS or SSS.

Used in: 8 · 10 · 14

O

OPERATION 7

Discover a congruence

Licensed by Thm. 2 (SAS) or Thm. 6 (SSS). Prove two triangles equal, then take the corresponding part. The most used of the nine.

Used in: 3 · 6 · 7 · 8 · 9 · 10 · 14

O

OPERATION 8

Add equals, subtract equals, or chain equals

Licensed by C.N. 1 · C.N. 2 · C.N. 3. Used constantly, and the most often left uncited.

Used in: 1 · 3 · 11 · 12 · 13



OPERATION 9

Whole greater than part

Licensed by C.N. 5. Every inequality, and nearly every contradiction. When stuck, try this.

Used in: 4 · 5 · 12 · 14 · 15

